

Dual Park–Ravani Interpolation of Rigid Motions: Acceleration-Field Continuity and Holonomic Hermite Repair

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Abstract

The Park–Ravani construction generates a twice continuously differentiable, frame-invariant spline on $SO(3)$ by exponentiating cubic canonical-coordinate polynomials. We show that the complete construction transfers, without a change of form, to the group of orthogonal dual tensors, a representation of rigid displacements. The transferred recurrence is stated compactly through the dual extension of the right Jacobian of the exponential map and its first Fréchet derivative. This yields interpolation of prescribed rigid poses and continuity of the body dual twist and its first derivative. Using the higher-order rigid-body kinematics of dual spatial twists, we then prove that the resulting curve has a continuous physical acceleration field, not merely a continuous quantity obtained by formally differentiating the dual part of a twist. We also distinguish algebraic dual transfer from temporal differential prolongation: their simultaneous first-order use takes place in a hyper-dual algebra, and interpolation of arbitrary prolonged nodal data need not be holonomic. A noncommuting three-pose example verifies the recurrence, all knot continuity statements, and dimensional covariance under a change from metres to millimetres. We define and analyse the first-order holonomy defect of a generic hyper-dual interpolant, exhibit an exact counterexample, and remove the defect by cubic or quintic Hermite interpolation in dual logarithmic coordinates.

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1 Introduction

Park and Ravani introduced an efficient invariant interpolation scheme on $SO(3)$ in which each segment is the exponential of a cubic polynomial in canonical coordinates [1]. The construction interpolates an ordered set of rotations, is invariant under changes of fixed and moving frames, and produces a C^2 rotation curve. Its relation to minimum angular acceleration is approximate in general and exact only in the special cases identified in the original paper; no stronger variational assertion is used here.

Invariant rotation and rigid-motion interpolation has also been developed through quaternion curves [2, 3], Lie-group versions of the de Casteljau algorithm [4], and interpolation or projection schemes on $SE(3)$ [5, 6]. Dual-quaternion blending and screw interpolation provide a further computational representation of rigid displacements [7]. These approaches establish the broader geometric setting, but they do not supply the dual-tensor continuation of the Park–Ravani recurrence or the acceleration-field and holonomy results proved below. Unlike projection or blending methods, the present construction retains the Park–Ravani recurrence in canonical coordinates and transfers it exactly to a dimensionally homogeneous dual-tensor representation; no projection back to $SE(3)$ is required.

Rigid displacement, however, possesses a natural dual description. An orthogonal dual tensor combines rotation and translation without placing dimensionless and length-valued entries in the same real matrix norm. This suggests a direct question: does the Park–Ravani

construction itself extend from rotations to rigid motions by dual continuation? The answer is yes. More precisely, every algebraic and analytic relation used by the construction extends to the dual algebra, provided the logarithm is evaluated in one admissible local branch [14]. The representation used here builds on dual-tensor formulations of rigid motion [8] and on the classical principle of transference [9, 10, 11]. Hyper-dual arithmetic is used only where the directional derivative of the Jacobian must be evaluated exactly, in the automatic-differentiation sense of [12, 13].

A second question is subtler. The dual part of the time derivative of a dual twist is often informally called a linear acceleration. In general this is incorrect. The derivative of the spatial dual twist determines a helicoidal field of reduced accelerations; the complete physical acceleration field is determined by the spatial dual twist together with its first derivative [15]. Consequently, the physically meaningful smoothness statement for a dual Park–Ravani spline must be proved at the level of the entire point-acceleration field.

The contribution is not a formal substitution of dual quantities into a real algorithm. The paper establishes: (i) an exact transfer theorem for the complete Park–Ravani construction on orthogonal dual tensors; (ii) a compact nonuniform-knot recurrence using the right Jacobian and its directional derivative; (iii) continuity of the complete physical acceleration field of every material point, rather than only of a formally differentiated dual component; (iv) a precise distinction between dual extension and temporal differential transformation, including an exact nonholonomy counterexample; and (v) minimal-degree cubic and quintic dual-logarithmic Hermite repairs for prescribed endpoint twist and acceleration-field data. Thus transfer, physical interpretation, and holonomic repair form a single closed argument.

Theorem 5.1 gives a precise algorithmic form of the classical principle of transference for the complete Park–Ravani construction. Its scope is deliberately algebraic: analytic identities transfer from real to dual data. Proposition 7.2 identifies the boundary of this statement. After an independent temporal nilpotent unit is introduced, coefficientwise algebraic transfer does not ensure that the resulting hyper-dual curve is the temporal jet of its base projection. Algebraic transference and temporal holonomy are therefore distinct properties.

2 Dual orthogonal tensors and rigid displacements

2.1 Dual vectors and tensors

Let

$$\mathbb{D} = \mathbb{R}[\varepsilon_0]/(\varepsilon_0^2)$$

be the algebra of dual numbers. Throughout the paper, an underline denotes a dual quantity:

$$\underline{x} = x + \varepsilon_0 x^\circ, \quad \underline{A} = A + \varepsilon_0 A^\circ.$$

The unit ε_0 is the rigid, or geometric, nilpotent unit. It is dimensionless as an algebraic generator; physical dimensions reside in the coefficients. For example, the real part of a dual rotation vector is angular, whereas its dual part has the dimension of length.

The transpose, scalar product, cross product, and tensor product are extended by dual bilinearity. In particular,

$$(x + \varepsilon_0 x^\circ) \times (y + \varepsilon_0 y^\circ) = x \times y + \varepsilon_0 (x \times y^\circ + x^\circ \times y).$$

Hence $\underline{x} \times \underline{x} = 0$.

Definition 2.1. *An orthogonal dual tensor is a dual tensor $\underline{R}$ satisfying*

$$\underline{R}^T \underline{R} = I, \quad \det \underline{R} = 1.$$

The resulting group is denoted by $SO(3, \mathbb{D})$.

Every rigid displacement (R, p) admits the representation

$$\underline{R} = (I + \varepsilon_0[p]_{\times})R = R + \varepsilon_0[p]_{\times}R, \quad R \in SO(3), \quad p \in \mathbb{R}^3. \quad (1)$$

Composition and inversion are ordinary dual-tensor multiplication and transpose. Thus $SO(3, \mathbb{D})$ is a dimensionally homogeneous realization of the rigid-displacement group.

The Lie algebra consists of dual skew tensors. We identify $[\underline{x}]_{\times}$ with the dual vector $\underline{x}$. Exponential and logarithm are defined by analytic continuation of their real tensor series. An *admissible logarithm branch* is a local inverse of the exponential selected around a dual logarithm $\underline{x}$ whose real rotation-vector norm is not a nonzero multiple of 2π [14]. The value $\|\text{Re } \underline{x}\| = \pi$ is admissible after fixing one of the two axis signs: this is a branch choice, not a singularity of the exponential Jacobian. Pure translation is included by the regular zero-angle continuation. On such a branch,

$$\exp[\underline{x}]_{\times} \in SO(3, \mathbb{D}), \quad \log \underline{R} = [\underline{x}]_{\times}.$$

2.2 Body and spatial dual twists

For a differentiable rigid motion $\underline{R}(t)$, define

$$[\underline{\omega}_b]_{\times} = \underline{R}^T \dot{\underline{R}}, \quad [\underline{\omega}_s]_{\times} = \dot{\underline{R}} \underline{R}^T. \quad (2)$$

The corresponding dual-vector relation is

$$\underline{\omega}_s = \underline{R} \underline{\omega}_b. \quad (3)$$

Differentiating and using $\dot{\underline{R}} \underline{x} = \underline{R}(\underline{\omega}_b \times \underline{x})$ gives

$$\dot{\underline{\omega}}_s = \underline{R} \dot{\underline{\omega}}_b + \underline{R}(\underline{\omega}_b \times \underline{\omega}_b) = \underline{R} \dot{\underline{\omega}}_b. \quad (4)$$

This cancellation will be essential at an interpolation knot.

3 Temporal differential transform

Let $\underline{x}(t)$ be a dual-valued differentiable function. Its first temporal differential transform is

$$\check{\underline{x}}(t) = \underline{x}(t) + \varepsilon \dot{\underline{x}}(t), \quad \varepsilon^2 = 0. \quad (5)$$

Here ε is distinct from the rigid dual unit ε_0 . The two units commute, so the global coefficient algebra is

$$\mathbb{H}_D = \mathbb{R}[\varepsilon_0, \varepsilon] / (\varepsilon_0^2, \varepsilon^2), \quad \varepsilon_0 \varepsilon \neq 0. \quad (6)$$

Indeed,

$$\check{\underline{x}} = \underline{x} + \varepsilon_0 \underline{x}^{\circ} + \varepsilon \dot{\underline{x}} + \varepsilon_0 \varepsilon \dot{\underline{x}}^{\circ}.$$

Thus the first differential transform of a dual quantity is hyper-dual. No hat notation is used at this level; it is reserved for higher-order extensions.

Because (5) is exact first-order automatic differentiation, analytic identities prolong functorially:

$$\underline{F}(\check{\underline{x}}) = F(\check{\underline{x}}) = F(\underline{x}) + \varepsilon DF(\underline{x})[\dot{\underline{x}}]. \quad (7)$$

This applies in particular to the exponential, the logarithm inside an admissible chart, and the Jacobian used below. Equation (7) must not be confused with holonomy of an interpolated hyper-dual curve: arbitrary transformed nodal data need not be the temporal transform of the curve obtained by projecting their base parts.

4 The dual Park–Ravani construction

Let $t_0 < \dots < t_N$ be knot times, $h_i = t_{i+1} - t_i$, and let $\underline{R}_i \in SO(3, \mathbb{D})$ be prescribed rigid poses. Choose one admissible logarithm branch for every relative displacement, fixing an axis sign when the real relative angle is π , and set

$$[\underline{s}_i]_{\times} = \log(\underline{R}_i^T \underline{R}_{i+1}). \quad (8)$$

On the normalized interval $u = (t - t_i)/h_i \in [0, 1]$, define

$$\underline{r}_i(u) = \underline{a}_i u^3 + \underline{b}_i u^2 + \underline{c}_i u, \quad (9)$$

$$\underline{R}(t) = \underline{R}_i \exp[\underline{r}_i(u)]_{\times}. \quad (10)$$

Endpoint interpolation is equivalent to

$$\underline{a}_i = \underline{s}_i - \underline{b}_i - \underline{c}_i. \quad (11)$$

For a dual vector $\underline{r}$ define the dual right Jacobian

$$\underline{J}(\underline{r}) = I - \frac{1 - \cos \underline{q}}{\underline{q}^2} [\underline{r}]_{\times} + \frac{\underline{q} - \sin \underline{q}}{\underline{q}^3} [\underline{r}]_{\times}^2, \quad \underline{q}^2 = \underline{r} \cdot \underline{r}, \quad (12)$$

with the regular analytic continuation at $\underline{q} = 0$. Under the convention (2), the body dual twist of (10) is

$$\underline{\omega}_b(t) = \frac{1}{h_i} \underline{J}(\underline{r}_i) \underline{r}_i'. \quad (13)$$

Its time derivative is

$$\dot{\underline{\omega}}_b(t) = \frac{1}{h_i^2} \{ D\underline{J}(\underline{r}_i) [\underline{r}_i'] \underline{r}_i' + \underline{J}(\underline{r}_i) \underline{r}_i'' \}. \quad (14)$$

Put

$$\underline{\tau}_i = 3\underline{a}_i + 2\underline{b}_i + \underline{c}_i, \quad \underline{\nu}_i = 6\underline{a}_i + 2\underline{b}_i. \quad (15)$$

These are the first two u -derivatives of $\underline{r}_i$ at $u = 1$.

Proposition 4.1 (Nonuniform dual Park–Ravani recurrence). *Assume that the initial body dual twist $\underline{\omega}_{b,0}$ and its derivative $\dot{\underline{\omega}}_{b,0}$ are prescribed. Set*

$$\underline{c}_0 = h_0 \underline{\omega}_{b,0}, \quad \underline{b}_0 = \frac{h_0^2}{2} \dot{\underline{\omega}}_{b,0}, \quad \underline{a}_0 = \underline{s}_0 - \underline{b}_0 - \underline{c}_0. \quad (16)$$

For $i = 0, \dots, N-2$, define successively

$$\underline{c}_{i+1} = \frac{h_{i+1}}{h_i} \underline{J}(\underline{s}_i) \underline{\tau}_i, \quad (17)$$

$$2\underline{b}_{i+1} = \frac{h_{i+1}^2}{h_i^2} \{ D\underline{J}(\underline{s}_i) [\underline{\tau}_i] \underline{\tau}_i + \underline{J}(\underline{s}_i) \underline{\nu}_i \}, \quad (18)$$

$$\underline{a}_{i+1} = \underline{s}_{i+1} - \underline{b}_{i+1} - \underline{c}_{i+1}. \quad (19)$$

Then (10) interpolates every $\underline{R}_i$, and its body dual twist and body dual-twist derivative are continuous at every internal knot.

Proof. At $u = 0$, $\underline{J}(0) = I$ and $D\underline{J}(0)[\underline{c}]\underline{c} = 0$. Equations (13)–(14) therefore give the initial data (16). At $u = 1$, $\underline{r}_i = \underline{s}_i$, $\underline{r}_i' = \underline{\tau}_i$, and $\underline{r}_i'' = \underline{\nu}_i$. The left limits of the twist and its derivative are consequently

$$\frac{1}{h_i} \underline{J}(\underline{s}_i) \underline{\tau}_i, \quad \frac{1}{h_i^2} \{ D\underline{J}(\underline{s}_i) [\underline{\tau}_i] \underline{\tau}_i + \underline{J}(\underline{s}_i) \underline{\nu}_i \}.$$

The corresponding right limits on segment $i+1$ are $\underline{c}_{i+1}/h_{i+1}$ and $2\underline{b}_{i+1}/h_{i+1}^2$. Their equality is precisely (17)–(18). Finally, (19) gives $\underline{r}_{i+1}(1) = \underline{s}_{i+1}$, hence pose interpolation. $\square$

5 Exact dual transfer

Write a complete set of real Park–Ravani input data as Y and denote the resulting real spline construction by $\mathcal{P}(Y)$. Its dual continuation is not a new independently postulated algorithm.

Theorem 5.1 (Dual transfer principle). *Let $Y + \varepsilon_0 Y^\circ$ be dual input data lying in a product of admissible logarithm charts. Then*

$$\mathcal{P}_{\mathbb{D}}(Y + \varepsilon_0 Y^\circ) = \mathcal{P}(Y) + \varepsilon_0 D\mathcal{P}(Y)[Y^\circ]. \quad (20)$$

In particular, every analytic identity used in the real Park–Ravani construction, including its recurrence and frame covariance, holds over the dual algebra with exactly the same form.

Proof. Addition, multiplication, contraction, and cross product extend by dual bilinearity. For every analytic scalar or tensor function F ,

$$F(X + \varepsilon_0 X^\circ) = F(X) + \varepsilon_0 DF(X)[X^\circ],$$

because $\varepsilon_0^2 = 0$. The Park–Ravani construction is a finite composition of these operations with exponential, local logarithm, and the analytic functions entering J . The chain rule therefore gives (20). Applying the same continuation to any real analytic identity proves the final assertion. $\square$

Remark 5.2. *The theorem is an exact first-order transfer in the rigid dual direction. It does not assert a variational minimum in a lexicographically ordered dual functional. Such a statement requires a separate definition and proof.*

Remark 5.3 (Forward propagation). *As in the original Park–Ravani construction, the recurrence propagates the initial twist data forward and imposes no terminal condition. Over long knot sequences, coefficient growth may therefore occur. The present paper establishes the exact transferred recurrence and its physical continuity properties, not a global conditioning result.*

6 Continuity of the physical acceleration field

Let the spatial dual twist and its derivative be decomposed as

$$\underline{\omega}_s = \omega + \varepsilon_0 v, \quad \underline{\dot{\omega}}_s = \dot{\omega} + \varepsilon_0 \dot{v}. \quad (21)$$

The vector $\dot{v}$ is the reduced acceleration of the instantaneous point of the rigid body passing through the origin. It is not, by itself, the physical linear acceleration of a fixed material point. For a spatial point with position vector ρ , define

$$\Phi_2 = [\dot{\omega}]_{\times} + [\omega]_{\times}^2, \quad (22)$$

$$a_2 = \dot{v} + \omega \times v, \quad (23)$$

$$a_\rho = a_2 + \Phi_2 \rho. \quad (24)$$

Equation (24) is the physical acceleration field of the rigid body [15].

Theorem 6.1 (Acceleration-field continuity). *The dual Park–Ravani spline generated by Proposition 4.1 has a continuous physical acceleration field at every internal knot. More explicitly, for each fixed knot time t_i , the left and right limits of a_ρ coincide for every $\rho \in \mathbb{R}^3$.*

Proof. Proposition 4.1 gives equality at the knot of the left and right limits of $\underline{R}$, $\underline{\omega}_b$, and $\underline{\dot{\omega}}_b$. Equations (3) and (4) imply equality of the corresponding spatial pair

$$(\underline{\omega}_s, \underline{\dot{\omega}}_s).$$

Taking real and dual parts in (21) yields identical left and right values of ω , v , $\dot{\omega}$, and $\dot{v}$. Therefore (22) and (23) give identical Φ_2 and a_2 . Substitution in (24) proves equality of a_ρ for every spatial point ρ . $\square$

Remark 6.2. *Continuity of the dual-twist derivative alone is not the appropriate physical interpretation. For $n = 2$, the pair $(\underline{\omega}_s, \underline{\dot{\omega}}_s)$ uniquely determines the complete physical acceleration field; the dual Park–Ravani recurrence makes this pair continuous.*

7 Holonomy and its Hermite repair

7.1 The first-order holonomy condition

Let a curve with coefficients in the hyper-dual algebra (6) be written as

$$\check{\underline{R}}(t) = \underline{R}_0(t) + \varepsilon \underline{R}_1(t). \quad (25)$$

Its base projection is $\pi_0 \check{\underline{R}} = \underline{R}_0$. If $\check{\underline{R}}$ is orthogonal over the hyper-dual algebra, then $\underline{R}_0 \in SO(3, \mathbb{D})$ and $\underline{R}_0^T \underline{R}_1$ is dual skew. This tangency condition is necessary but does not say that the ε coefficient is the time derivative of the base curve.

Definition 7.1 (First-order holonomy). *The curve (25) is temporally holonomic if*

$$\boxed{\underline{R}_1(t) = \dot{\underline{R}}_0(t)} \quad (26)$$

throughout its interval. Equivalently, $\check{\underline{R}} = \underline{R}_0 + \varepsilon \dot{\underline{R}}_0$ is the temporal differential transform of its base projection.

Define the tensor defect and its body-trivialized dual vector by

$$\underline{\Delta}(t) = \underline{R}_1(t) - \dot{\underline{R}}_0(t), \quad (27)$$

$$[\delta_b(t)]_\times = \underline{R}_0(t)^T \underline{\Delta}(t). \quad (28)$$

For an orthogonal hyper-dual curve, the right-hand side of (28) is dual skew. Holonomy is equivalent to $\underline{\delta}_b \equiv 0$. Under a constant change of fixed frame the body defect is unchanged; under a constant change of moving frame it transforms by the corresponding dual orthogonal action.

Proposition 7.2 (Algebraic interpolation does not imply holonomy). *Applying the dual Park–Ravani formulas coefficientwise to individually prolonged nodal data does not, in general, produce the temporal differential transform of the spline obtained from the base data.*

Proof. It is sufficient to work in a one-parameter, hence commutative, subgroup. Let $u = t \in [0, 1]$ and fix a nonzero dual vector $\underline{e}$. Write

$$\underline{R}_0(u) = \exp[x(u)\underline{e}]_\times.$$

Take base endpoint coordinates $x(0) = 0$, $x(1) = 1$ and zero initial first and second derivatives. The Park–Ravani cubic is

$$x(u) = u^3. \quad (29)$$

Attach zero temporal coefficients to both endpoint coordinates and to the initial derivative data. Coefficientwise application of the same construction gives

$$\check{x}(u) = u^3 + \varepsilon 0. \quad (30)$$

The temporal differential transform of the base interpolant is instead

$$x(u) + \varepsilon x'(u) = u^3 + 3\varepsilon u^2. \quad (31)$$

Thus the scalar defect is $-3u^2$, and the tensor defect is nonzero for every $u \in (0, 1]$. The failure already occurs in a commutative subgroup, so noncommutativity is not its cause. $\square$

Remark 7.3. *Each nodal object in the counterexample is individually a legitimate temporal transform. The obstruction is global: the independently assigned nodal coefficients are not the jet of the base spline selected by the interpolation algorithm. Functorial prolongation (7) and holonomy are therefore distinct statements.*

7.2 Cubic dual-logarithmic Hermite interpolation

Consider one interval $[t_i, t_{i+1}]$, put $u = (t - t_i)/h_i$, and let $\underline{s}_i$ be given by (8), with $\|\operatorname{Re} \underline{s}_i\| \notin 2\pi\mathbb{Z} \setminus \{0\}$. Suppose the body dual twists $\underline{\omega}_{b,i}$ and $\underline{\omega}_{b,i+1}$ are prescribed. Define

$$\underline{d}_0 = h_i \underline{\omega}_{b,i}, \quad \underline{d}_1 = h_i J(\underline{s}_i)^{-1} \underline{\omega}_{b,i+1}. \quad (32)$$

The inverse in (32) is well defined in the selected admissible logarithm chart. Indeed, if $\underline{J} = J + \varepsilon_0 J^\circ$ and the real Jacobian J is nonsingular, then

$$\underline{J}^{-1} = J^{-1} - \varepsilon_0 J^{-1} J^\circ J^{-1}. \quad (33)$$

The cubic dual-vector polynomial

$$\begin{aligned} \underline{r}_H(u) &= (3u^2 - 2u^3)\underline{s}_i + (u^3 - 2u^2 + u)\underline{d}_0 \\ &\quad + (u^3 - u^2)\underline{d}_1 \end{aligned} \quad (34)$$

satisfies

$$\underline{r}_H(0) = 0, \quad \underline{r}_H(1) = \underline{s}_i, \quad \underline{r}'_H(0) = \underline{d}_0, \quad \underline{r}'_H(1) = \underline{d}_1.$$

Theorem 7.4 (Holonomic cubic repair). *The curve*

$$\underline{R}_H(t) = \underline{R}_i \exp[\underline{r}_H(u)]_\times \quad (35)$$

interpolates the two dual poses and the two prescribed body dual twists. Its temporal differential transform

$$\check{\underline{R}}_H(t) = \underline{R}_H(t) + \varepsilon \dot{\underline{R}}_H(t) \quad (36)$$

is identically holonomic.

Proof. The endpoint values of $\underline{r}_H$ give the two poses. At $u = 0$, $\underline{J}(0) = I$, and (13) gives $\underline{\omega}_b(t_i) = \underline{d}_0/h_i = \underline{\omega}_{b,i}$. At $u = 1$ it gives

$$\underline{\omega}_b(t_{i+1}) = h_i^{-1} \underline{J}(\underline{s}_i) \underline{d}_1 = \underline{\omega}_{b,i+1}.$$

Equation (36) satisfies (26) by construction, so its defect (27) vanishes identically. $\square$

For the scalar data in Proposition 7.2, with zero endpoint velocities, the repaired base coordinate and its prolongation are

$$x_H(u) = 3u^2 - 2u^3, \quad \check{x}_H(u) = 3u^2 - 2u^3 + \varepsilon(6u - 6u^2).$$

They have zero defect and match both prescribed endpoint velocities.

7.3 Quintic repair for prescribed accelerations

If the body dual-twist derivatives are prescribed at both endpoints, set

$$\underline{e}_0 = h_i^2 \dot{\underline{\omega}}_{b,i}, \quad (37)$$

$$\underline{e}_1 = \underline{J}(\underline{s}_i)^{-1} \{ h_i^2 \dot{\underline{\omega}}_{b,i+1} - D\underline{J}(\underline{s}_i)[\underline{d}_1] \underline{d}_1 \}. \quad (38)$$

Introduce the quintic Hermite basis

$$\begin{aligned} H_{01} &= 10u^3 - 15u^4 + 6u^5, \\ H_{10} &= u - 6u^3 + 8u^4 - 3u^5, \\ H_{11} &= -4u^3 + 7u^4 - 3u^5, \\ H_{20} &= \frac{1}{2}(u^2 - 3u^3 + 3u^4 - u^5), \\ H_{21} &= \frac{1}{2}(u^3 - 2u^4 + u^5), \end{aligned}$$

and define

$$\underline{r}_Q(u) = H_{01}\underline{s}_i + H_{10}\underline{d}_0 + H_{11}\underline{d}_1 + H_{20}\underline{e}_0 + H_{21}\underline{e}_1. \quad (39)$$

Theorem 7.5 (Holonomic quintic repair). *The curve*

$$\underline{R}_Q(t) = \underline{R}_i \exp[\underline{r}_Q(u)]_\times$$

interpolates the endpoint dual poses, body dual twists, and body dual-twist derivatives. Its temporal differential transform is identically holonomic. Consequently, the prescribed physical acceleration fields are attained at both endpoints.

Proof. The quintic basis gives

$$(\underline{r}_Q, \underline{r}'_Q, \underline{r}''_Q)(0) = (0, \underline{d}_0, \underline{e}_0), \quad (\underline{r}_Q, \underline{r}'_Q, \underline{r}''_Q)(1) = (\underline{s}_i, \underline{d}_1, \underline{e}_1).$$

Equations (13) and (14), together with (32)–(38), give the prescribed endpoint twists and derivatives. Temporal prolongation of this single base curve makes (26) an identity. The endpoint physical acceleration fields follow from (24). $\square$

Remark 7.6. *The cubic and quintic constructions solve different interpolation problems. The cubic matches pose and twist. The quintic matches pose, twist, and twist derivative, hence the complete physical acceleration field. The latter is not claimed to be the unique possible holonomic repair; it is the minimal-degree polynomial in one logarithm chart satisfying these six endpoint conditions.*

8 Numerical verification

8.1 Data and implementation

We consider three prescribed rigid poses at the nonuniform knot times

$$t_0 = 0, \quad t_1 = 1, \quad t_2 = 2.5, \quad h_0 = 1, \quad h_1 = 1.5.$$

The real rotation vectors q_i , in radians, and translations p_i , in metres, are

i	q_{i1}	q_{i2}	q_{i3}	p_{i1}	p_{i2}	p_{i3}
0	0.08	−0.04	0.06	0.05	−0.02	0.03
1	0.38	−0.22	0.29	0.42	−0.16	0.27
2	−0.18	0.46	0.34	0.83	0.31	−0.12

(40)

and the poses are formed as

$$R_i = \exp[q_i]_{\times}, \quad \underline{R}_i = (I + \varepsilon_0[p_i]_{\times})R_i.$$

The initial body dual twist and its derivative are prescribed as

$$\underline{\omega}_{b,0} = (0.24, -0.13, 0.19) + \varepsilon_0(0.18, 0.07, -0.11), \quad (41)$$

$$\underline{\dot{\omega}}_{b,0} = (0.11, 0.08, -0.06) + \varepsilon_0(-0.09, 0.14, 0.05). \quad (42)$$

Angular components use seconds as the time unit; the dual components carry the corresponding metre-based dimensions.

The two relative logarithms obtained from (8) are

$$\begin{aligned} \underline{s}_0 &= (0.2992674465, -0.1796622622, 0.2312136665) \\ &\quad + \varepsilon_0(0.3750816019, -0.1514565323, 0.2248304621), \end{aligned} \quad (43)$$

$$\begin{aligned} \underline{s}_1 &= (-0.4427966639, 0.7577255960, -0.0168351043) \\ &\quad + \varepsilon_0(0.5945254558, 0.2816054523, -0.3798654220). \end{aligned} \quad (44)$$

These rotations do not share a common axis, so the test is genuinely noncommutative.

All analytic functions of dual tensors are evaluated by their exact dual extensions. In particular, the quantity

$$DJ(\underline{s}_i)[\underline{\tau}_i]\underline{\tau}_i$$

in (18) is computed by a second nilpotent unit ε : the coefficient of ε in $J(\underline{s}_i + \varepsilon \underline{\tau}_i)\underline{\tau}_i$ is extracted. Thus the calculation uses the hyper-dual algebra (6) and no finite-difference approximation.

The Python scripts reproducing every numerical value reported in this section are provided as ancillary files. A single driver script executes the metre and millimetre experiments, the dimensional-covariance check, and the independent cubic and quintic Hermite verifications.

8.2 Knot-continuity residuals

The compact recurrence (17)–(18) was also evaluated after expanding DJ into the trigonometric Park–Ravani formula. Table 1 reports Euclidean norms obtained by stacking the real and dual vector components, or Frobenius norms applied separately to the real and dual tensor parts and then combining their numerical entries. They are diagnostics in the fixed radian–metre–second convention, not intrinsic norms on the dual Lie algebra. Dimensional covariance is assessed separately in Section 8.3.

Most residuals in Tables 1–2 verify mutually equivalent algebraic representations within the implementation. The finite-difference differentiation of the complete dual-tensor curves in Section 8.4 provides the independent curve-level check.

Table 1: Residuals at the internal knot t_1 .

Verification	Residual
Compact versus expanded recurrence	8.33×10^{-17}
Dual pose interpolation	8.05×10^{-16}
Body dual twist continuity	1.36×10^{-16}
Body dual-twist derivative continuity	2.29×10^{-16}
Spatial dual twist continuity	2.00×10^{-16}
Spatial dual-twist derivative continuity	2.29×10^{-16}
Maximum point-acceleration-field residual	3.14×10^{-16}

For an explicit field-level check, (24) was evaluated at four points: p_1 and the three points obtained from p_1 by offsets $0.25e_1$, $0.20e_2$, and $0.18e_3$ metres. The common acceleration values at the knot are shown in Table 2. The maximum norm of the left-minus-right difference is the last residual of Table 1.

Table 2: Physical point accelerations at t_1 in m s^{-2} .

Point	a_x	a_y	a_z
p_1	1.336338	-1.792470	1.773506
$p_1 + 0.25e_1$	1.284059	-1.754410	1.928334
$p_1 + 0.20e_2$	1.256146	-1.840993	1.788991
$p_1 + 0.18e_3$	1.267878	-1.844018	1.728332

8.3 Dimensional covariance

The entire experiment was repeated after replacing metres by millimetres. Every translational datum, dual coefficient, and point offset was therefore multiplied by 10^3 , whereas angles and times were unchanged. The real parts of both relative logarithms remained identical. Their dual parts scaled by 10^3 with a maximum absolute discrepancy of 2.27×10^{-13} mm. The computed physical accelerations also scaled by 10^3 ; the maximum relative scaling error was

$$8.25 \times 10^{-16}. \quad (45)$$

The largest knot residual in the millimetre computation was 5.57×10^{-13} , consistent with the corresponding change in the magnitude of the length-valued components.

This test is not a claim that a numerical norm is independent of units. Rather, it verifies the correct covariance of each homogeneous component: angular quantities remain unchanged and length-valued quantities acquire the same scale factor as the unit of length. No norm combines a dimensionless rotation matrix with a translation column.

8.4 Independent curve-level check of the Hermite construction

The cubic and quintic curves were constructed explicitly for a separate dual rigid displacement with independently prescribed twists and twist derivatives at both ends. Their dual tensor exponentials were then differentiated by five-point, fourth-order finite-difference stencils. This verification is independent of the J and DJ formulas used to generate the Hermite data. The resulting pose, twist, and twist-derivative residuals are reported in Table 3. The larger derivative residuals reflect finite-difference truncation and cancellation rather than failure of the exact endpoint identities.

Table 3: Independent curve-level verification of the Hermite interpolants.

Residual	Initial endpoint	Final endpoint
Cubic pose	0	8.05×10^{-16}
Cubic body twist	7.14×10^{-13}	4.04×10^{-12}
Quintic pose	0	8.05×10^{-16}
Quintic body twist	1.59×10^{-13}	1.73×10^{-12}
Quintic body-twist derivative	1.08×10^{-10}	6.32×10^{-9}

The earlier algebraic round-trip residuals, retained in the software as auxiliary implementation checks, remain between 2.40×10^{-17} and 3.40×10^{-17} . Direct evaluation of the

five quintic basis polynomials and their first two derivatives verifies all endpoint Kronecker conditions exactly. For the scalar counterexample, the maximum defect is 3, whereas the repaired prolongation has identically zero defect.

9 Conclusion and next steps

The Park–Ravani construction transfers exactly from $SO(3)$ to orthogonal dual tensors and therefore to rigid displacements. In compact form, its nonuniform recurrence is governed by the dual right Jacobian and one directional derivative. The transferred spline interpolates the prescribed poses and preserves both the dual twist and its first derivative across knots. The latter statement has been translated into its correct physical content: the entire rigid-body acceleration field is continuous.

The noncommuting three-pose example verifies the compact recurrence, twist continuity, acceleration-field continuity, and dimensional covariance under a metre-to-millimetre change. The computations use exact hyper-dual directional differentiation rather than finite differences. The first-order holonomy problem has also been closed: coefficientwise interpolation of independently prolonged nodal data is generically nonholonomic, while cubic and quintic dual-logarithmic Hermite curves provide exact repairs for endpoint twist and acceleration-field data, respectively. Higher temporal orders, their contact defects, and holonomic Hermite interpolation of general rigid-motion jets are left for future work.

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